JIA-PENG ZHANG 张嘉鹏

zjp24@mails.tsinghua.edu.cn · zjp-shadow.github.io · zjp-shadow · Google Scholar

EDUCATION

Tsinghua University, Beijing, China — Sep. 2024 – Present

Ph.D. Student in Department of Computer Science and Technology

- **GPA:** 3.95 **Rank:** 27/158
- **Advisor:** Prof. Shi-Min Hu (Institute of Human-Computer Interaction and Media Integration)
- **Research Focus:** 3D Generation, Animation, Graphics Simulation & Embodied AI, and AI Infra.

Tsinghua University, Beijing, China — Sep. 2020 – Jul. 2024

B.S. in Information and Computing Science, Zhili College

- **Rank:** 21/30

SELECTED PUBLICATIONS

- **SkinTokens: A Learned Compact Representation for Unified Autoregressive Rigging**
  Jia-Peng Zhang, Cheng-Feng Pu, Meng-Hao Guo, Yan-Pei Cao, Shi-Min Hu
  ACM Transactions on Graphics (TOG), 2026 (zjp-shadow.github.io/works/SkinTokens/)
  Introduced *SkinTokens*, a learned discrete representation for skinning weights. By framing skinning as a sequence prediction task, the unified *TokenRig* framework jointly models skeletons and skin deformations, achieving substantial improvements in rigorous out-of-distribution geometry evaluations.
- **One Model to Rig Them All: Diverse Skeleton Rigging with UniRig**
  Jia-Peng Zhang, Cheng-Feng Pu, Meng-Hao Guo, Yan-Pei Cao, Shi-Min Hu
  ACM Transactions on Graphics (TOG) (***SIGGRAPH 2025***) (zjp-shadow.github.io/works/UniRig/)
  Proposed UniRig, a unified framework for automated skeletal rigging leveraging large autoregressive models and a novel *Skeleton Tree Tokenization* method. Trained on *Rig-XL* (a new large-scale dataset), it efficiently generates topologically valid structures, far outperforming existing state-of-the-art tools across diverse generic models.
- **CharacterGen: Efficient 3D Character Generation from Single Images with Multi-View Pose Canonicalization**
  Hao-Yang Peng, **Jia-Peng Zhang**, Meng-Hao Guo, Yan-Pei Cao, Shi-Min Hu
  ACM Transactions on Graphics (TOG) (***SIGGRAPH 2024***), 43(4): 1–13, 2024 (charactergen.github.io/)
  Presented a highly efficient framework for lifting single images into pose-unified 3D character meshes. It utilizes an image-conditioned multi-view diffusion model for pose calibration and a transformer-based sparse-view network, producing geometrically and texturally robust outputs ready for animation.
- **Transferring the Intelligence of VLMs to Robotic Control**
  Meng-Hao Guo, Zhe-Han Mo, Jia-Jun Wang, Yi Zhang, Kejin Wang, Yi-Xuan Deng, **Jia-Peng Zhang**, Yongming Rao, Shi-Min Hu
  arXiv preprint arXiv:2609.22966, 2026 (robodawn.top/)
  Introduced *RoboDawn*, a human-intuitive interface that lets an agentic VLM control robots in a closed loop through discrete translation, rotation and gripper commands, grounded by a few in-context demonstrations. Without task-specific robot training it achieves state-of-the-art success rates on RoboTwin 2.0 C2R and RoboDojo, and transfers to a real Franka arm.
- **TopoCap: Learning Topology-Agnostic Motion Priors for Monocular Video-to-Animation**
  Cheng-Feng Pu, **Jia-Peng Zhang**, Meng-Hao Guo, Yan-Pei Cao, Shi-Min Hu
  SIGGRAPH 2026 *Conference Papers* (arxiv.org/abs/2606.12153)
  Introduced TopoCap, the first unified framework that extracts motion from monocular video and retargets it onto characters with arbitrary, unseen skeletal topologies, from bipeds to hexapods and inanimate objects, without test-time optimization. A Graph CVAE learns a topology-agnostic motion manifold, and conditional flow matching predicts its codes from video; trained on *Mobjaverse*, a new dataset with over 5,000 skeletal topologies and 2 million frames.

Open Source Projects

UniRig & SkinTokens (github.com/VAST-AI-Research/UniRig, SkinTokens) 2200+ Stars

Project Maintainer & Core Contributor

- Maintain the official open-source implementations of UniRig (SIGGRAPH 2025, 1.8k+ stars) and SkinTokens (TOG 2026, 450+ stars), distributing unified pre-trained skeleton and skinning models that rig multi-genre meshes, from functional humans to fictional inorganic artifacts.
- Released the curated *Rig-XL* dataset and the unified *TokenRig* pipeline that predicts skeletons and skinning weights in a single autoregressive model, enriching the broader 3D automation community.

CharacterGen (github.com/zjp-shadow/CharacterGen) 830+ Stars

Project Maintainer & Core Contributor

- Designed and actively maintained the rendering and generation inference tools driving single-image 3D lifting, resolving pose ambiguity with canonical "A-pose" generation.
- Open-sourced multi-view diffusion synthesis and detailed texture back-projection, widely adopted by 3D anime avatar creators.

Tsinghua Jittor Framework (github.com/Jittor/Jittor, JittorInfer) 3200+ Stars

Core Member & Head of JittorInfer Framework

- Engineered **JittorInfer**, a high-performance C++ LLM inference framework adapted to domestic AI hardware such as Huawei Ascend processors.
- Spearheaded system-level computing abstractions, including CPU multi-stage operator fusion and end-to-end resource execution scheduling, which brought large throughput improvements when serving frontier models like the **DeepSeek** lineup compared to Ascend-vLLM.

Honors and Awards

Rising Star Pioneer Award, Huawei Kunpeng & Ascend Developer Conference (KADC) 2026

Comprehensive Scholarship for Ph.D. Students, Tsinghua University 2025

Silver Medal, National Olympiad in Informatics (NOI) 2019

Gold Medal, China Team Selection for Informatics (CTS) 2019

Internship Experience

Tencent Hunyuan 2026

Research Intern, Video Generation (Tsinghua Summer Social Practice)

- Studied how the metrics of video VAEs relate to the generation performance of flow-based video models.

Tencent 2024

Research Intern

- Conducted research on 3D multi-modal understanding.

Huawei 2023

Software Engineering Intern

- Developed and implemented system-level optimizations for sparse matrix multiplication targeting the SME (Scalable Matrix Extension) instruction set on Kunpeng chips.